Exam Date & Time: 29-Nov-2023 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOVEMBER-DECEMBER 2023

OPERATING SYSTEMS [DSE 3153]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) A) Consider a computer system running 4 processes: DocumentEditor (DE), MusicPlayer (MP), PDFViewer (PV), and BackupUtility (BU). The total execution times for these processes are 30, 20, 10, and 10 units, respectively. DE, MP, and PV start simultaneously, whereas BU starts 15 units of time after DE.

The breakdown of each process's activity in terms of computation and I/O is as follows:

- DocumentEditor (DE): Spends the first 10% of its execution time processing user edits (computation), then it spends 20% of execution time on saving documents to disk (Disk I/O), and the remaining time in computation, probably finalizing formatting and other editing tasks.
- MusicPlayer (MP): Spends the first 5% of its execution time processing user commands (computation), then it spends 60% of execution time on reading music files from the disk (Disk I/O), and the remaining time in computation for tasks such as sound equalization or adjustments.
- PDFViewer (PV): Spends the first 30% of its execution time rendering PDF pages (computation), the next 10% saving or reading PDF annotations from the disk (Disk I/O), and the remaining time in computation, likely for PDF page layout recalculations and other tasks. (5)
- BackupUtility (BU): Spends the first 70% of its time copying files for backup (computation), then it spends 20% of execution time on compressing backup files (Disk I/O), and the remaining 10% in computation, likely finalizing backup logs and updating internal statuses.

Assuming that the operating system uses a preemptive shortest job first scheduling algorithm for CPU scheduling, answer the following questions:

a) Construct the Gantt chart for this scenario.

b) Compute the average waiting time and average turnaround time.

c) Compute the percentage of time the CPU remains idle?

d) Compute the total number of context switches for this scenario.

B) Provide a detailed explanation of the various states a process goes through in an operating system and illustrate these states with a comprehensive diagram. Include descriptions of transitions between states and the events that trigger these transitions (3)

C) Explain why interrupt and dispatch latency times must be bounded in a hard real-time system. (2)

2) Consider a banking system that maintains an account balance with two functions: deposit (amount) and withdraw (amount). These two functions are passed the amount that is to be deposited or withdrawn from the bank account balance. Assume that a husband and wife

A) share a bank account. Concurrently, the husband calls the withdraw() function and the wife calls deposit(). Analyse how a race condition is possible and what might be done to prevent the race condition from occurring. (5)

B) Outline the differences and similarities among the three multithreading models: Many-to-One, One-to-One, and Many-to-Many. Highlight how each model handles the mapping of user-level threads to kernel-level threads and the implications of these approaches on performance and resource utilization. (3)

C) Consider the Resource Allocation Graph (RAG) given in Figure 2C. Based on the graph, answer the following:

a) Interpret the status of the Processes P1 and P2 in the graph.

b) Analyse the RAG and list out the scenarios which could lead to a potential deadlock. (2)

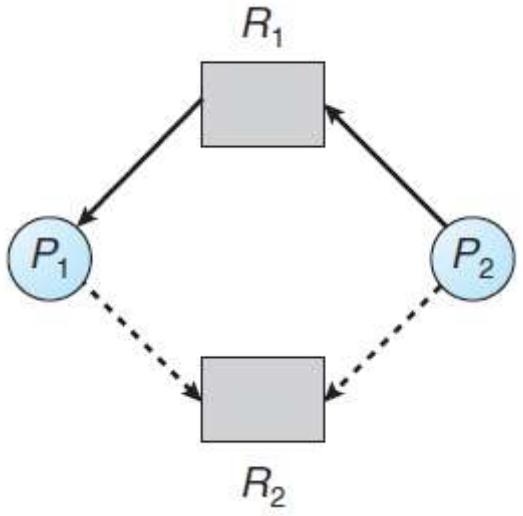


Figure 2C

3) Consider an operating system running five processes: Chrome, Photoshop, Spotify, Word, and System Update. The system has three types of resources: File Handles, Network Bandwidth, and Disk I/O Bandwidth. The total available units for these resources are 10

A) units of File Handles, 5 units of Network Bandwidth, and 7 units of Disk I/O Bandwidth, respectively. The maximum resources required by each process and the currently allocated resources are as follows: (5)

Process	Max File Handles	Max Network Bandwidth	Max Disk I/O	Allocated File Handles	Allocated Network Bandwidth	Allocated Disk I/O
Chrome	7	5	3	0	1	0
Photoshop	3	2	2	2	0	0
Spotify	9	0	2	3	0	2
Word	2	2	2	0	0	2
System Update	4	3	3	0	1	1

Assume Photoshop requests an additional allocation of 1 unit of File Handles, 0 units of Network Bandwidth, and 1 unit of Disk I/O Bandwidth. Determine if this request should be granted based on the current state of the system and the Banker's algorithm. Using the given initial allocation and maximum resources for each process, compute whether the system is in a safe state and provide the sequence in which the processes can complete their execution.

B) Consider a processor does not have a TSL (Test and Set lock) instruction, but does have an instruction "SWAP REG MEM" to swap the content of a register (REG) and a memory word (MEM) in a single indivisible (atomic) action. Prove that you can utilize SWAP to ensure mutual exclusion. Propose the entry_section and exit_section code modules. (3)

C) Is it possible to have a deadlock involving only one single-threaded process? Justify your answer. (2)

4) Consider a computer system with a 4 KB page size and 16 KB of physical memory. Initially, the system's memory contains four pages: A, B, C, and D, in that order. The system receives the following sequence of memory access requests: A, B, E, C, D, F, B, C, A, D.

A) Each memory access is for one page, and the system can only hold four pages in memory at a time.

Determine the sequence of page replacements, the final configuration of pages in the memory and the total number of page faults for the following page replacement algorithms: (5)

a) Second Chance,

b) Optimal,

c) Least Recently Used (stack implementation).

Identify which algorithm results in the minimum number of page faults.

B) With the help of a neat diagram, explain the address translation mechanism in a system which uses segmentation to allocate physical memory to the user programs. (3)

C) Differentiate between internal and external fragmentation. Also explain a solution to external fragmentation. (2)

5) (5)

A) Consider a disk with 200 tracks and the queue has random requests from different processes in the order: **55, 58, 39, 18, 90, 160, 150, 38, 184**. Initially disk arm is at 100. Find the Average Seek length using FIFO, SSTF, SCAN C-SCAN and LOOK algorithm. For SCAN, C-SCAN and LOOK assume that the disk arm is supposed to travel "towards the larger number".

B) Explain Belady's anomaly in the context of page replacement algorithms. Illustrate your answer by considering the following sequence of memory access requests:

0, 1, 5, 3, 0, 1, 4, 0, 1, 5, 3, 4, in that order, where each number indicates a request for a page number. Consider two scenarios where physical memory is divided into 3-frames and 4-frames, respectively. (3)

C) Consider a logical address space of 64 pages of 1,024 words each, mapped onto a physical memory of 32 frames.

a) How many bits are there in the logical address? (2)

b) How many bits are there in the physical address?

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